

MULTI-PRODUCTION OF UNIVERSES BY FIRST-ORDER PHASE TRANSITION OF A VACUUM

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Gauge theories with spontaneously broken symmetries give rise to a cosmological phase transition of a vacuum. We show that if the phase transition is strongly of first order, such gauge theories combined with general relativity yield a surprising prediction; although the Creator might have made a unitary universe, many mini-universes are produced sequentially afterward as a result of the phase transition.

The first-order phase transition of a vacuum has been a subject of great interest because of its notable implications to cosmology [1–8]. Since the universe is expected to expand exponentially during the period of the phase transition, the existence of this period makes it possible to form the seeds of galaxies [1], to explain the homogeneity of our universe [1–3], and to suppress the over-production of monopoles [5–7]. Furthermore, a matter–antimatter domain structure of the universe may be realized without observational conflicts [1,9].

The first-order phase transition is triggered by the nucleation of true vacuum bubbles [10]. At the early stage of the phase transition, these bubbles are isolated from each other, since their volume fraction to the background false vacuum is sufficiently small. At the late stage, however, they begin to overlap and finally cover most of the space. Thus false vacuum domains surrounded by true vacuum appear here and there. Now a paradoxical situation arises: Since a trapped false vacuum domain still expands exponentially, the area of the boundary surface of this region keeps increasing, while the very same surface is pushed inward at the light velocity due to the expansion of surrounding bubbles. Noting that, besides such a situation, there exists a high contrast between the energy densities of true and false vacua, one surely expects the universe to have a quite intricate space–time structure

during this stage. Let us first briefly discuss the space–time structure at this stage and show that a false vacuum region surrounded by true vacuum bubbles becomes eventually a black hole or a wormhole (for detail, see ref. [11]).

We assume that the radiation energy density is negligible compared to that of false vacuum, ρ_v , which is constant with time, and that the energy density inside true vacuum bubbles is completely vanishing. The metric of the universe before the nucleation of bubbles is then written as

$$ds^2 = -d\tau^2 + R^2(\tau)(d\chi^2 + \chi^2 d\Omega^2), \quad (1)$$

with the cosmic scale factor $R(\tau)$ given by

$$R(\tau) = \alpha_* l \exp[(\tau - \tau_*)/l], \quad (2)$$

where l is the characteristic length of the present model;

$$l \equiv (8\pi G\rho_v/3)^{-1/2}, \quad (3)$$

and α_* is a non-dimensional numerical factor which represents the scale of the universe at $\tau = \tau_*$. The cosmic time τ_* denotes the ignition time of the phase transition and we presume it to satisfy the relation $\tau_* \gg l$. As a model for spacetime with a trapped region of false vacuum, we adopt a spherically symmetric model, in which an infinite number of bubbles are nucleated on a sphere of radius $\chi = \chi_0$ simultaneously at time τ

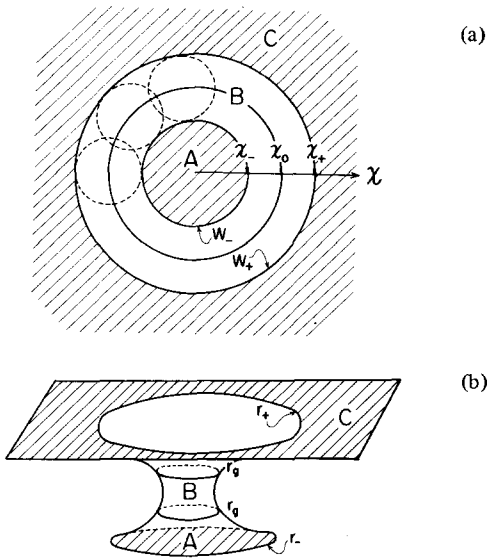


Fig. 1.

$= \tau_0 (> \tau_*)$, and they grow with walls expanding isotropically at the speed of light, forming a spherical shell-like region (see fig. 1a). From causality, the geometries of the false vacuum regions A and C are unaffected by the existence of the bubbles and still described by eq. (1). Meanwhile, the shell-like true vacuum region B is found to be described by the Schwarzschild geometry from Birkoff's theorem [12];

$$ds_B^2 = -(1 - r_g/r) dt^2 + (1 - r_g/r)^{-1} dr^2 + r^2 d\Omega^2. \quad (4)$$

The corresponding Schwarzschild mass M and radius r_g are given by

$$M = \frac{4}{3} \pi \rho_v [R(\tau_0) \chi_0]^3 = (\alpha \chi_0)^3 (4\pi \rho_v l^3 / 3) \quad (5)$$

and

$$r_g = 2GM = (\alpha \chi_0)^3 l, \quad \alpha \equiv \alpha_* \exp[(\tau_0 - \tau_*)/l]. \quad (6)$$

With the expansion of the region B, the coordinate radii $\chi_{\pm}$ of the outer and the inner walls $W_{\pm}$ change as

$$\begin{aligned} \chi_{\pm}(\tau) &= \chi_0 \pm \int_{\tau_0}^{\tau} dt'/R(t') \\ &= \chi_0 \pm \alpha^{-1} \{1 - \exp[-(\tau - \tau_0)/l]\}, \end{aligned} \quad (7)$$

so that their proper radii are given as

$$r_{\pm}(\tau) = R(\tau) \chi_{\pm}(\tau) = l \{(\alpha \chi_0 \pm 1) \exp[(\tau - \tau_0)/l] \mp 1\}. \quad (8)$$

If $\chi_0 < \alpha^{-1}$, r_- is initially greater than r_g , but decreases monotonically and vanishes within a finite time. Thus the inner false vacuum region eventually vanishes and a black hole of mass M remains in this case. On the other hand, if $\chi_0 > \alpha^{-1}$, r_- is initially smaller than r_g , but increases monotonically (and becomes infinite). This means that the inner region A remains as an ever expanding de Sitter-like sub-universe connected with the outer universe C by an Einstein-Rosen (wormhole) bridge B (see fig. 1b).

In reality, the trapped false vacuum regions are highly asymmetric and new bubbles are further nucleated therein. However, the essential feature of the above model is expected to be still applicable to the actual situation, provided that the nucleation rate is sufficiently small. The very reason why black holes and wormholes are created is that false vacuum domains trapped by true vacuum are formed. In case the relative size of a trapped domain to the surrounding bubbles is small when it is formed, the domain wall continues to contract at light speed and the energy released by the phase transition may well be concentrated in a very small region. Hence a black hole is generally created regardless of the initial shape of domain. On the other hand, in case the relative domain size is large, the domain wall continues to expand to infinity and consequently the universe attains to have two asymptotic (flat) regions. This anomalous structure occurs independent of the domain shape and strongly indicates the general formation of wormhole-bridge structure, though it is not exactly proved yet. It should be noticed that the nucleation rate of bubbles must be smaller than the characteristic value l^{-4} in order that the trapped domain can expand sufficiently without being destroyed by new bubbles nucleated therein. Thus one naturally expects these wormholes (and black holes) to play an important role in the subsequent evolution of the universe.

Let us now consider the creation rate and the fate of these wormholes. For simplicity, we assume the nucleation rate of bubbles per unit time and unit volume be independent of time and denote it by νl^{-4} . This assumption holds when the universe cools rapidly so that the nucleation by thermal fluctuations is negligible, and the nucleation is due only to quantum fluctuations which are independent of time. Then the volume fraction of the space occupied by false vacuum, $u(\tau)$, is given by [6,7]

$$u(\tau) = \exp \left\{ - \int_{\tau_*}^{\tau} \frac{4}{3} \pi \nu l^{-4} R^3(t) \chi^3(t, \tau) dt \right\}, \quad (9)$$

where $\chi(t, \tau) = \int_t^{\tau} dt' / R(t')$, and the number density of bubbles $n_B(\tau)$ is expressed as

$$n_B(\tau) = \nu u(\tau) / 3l^3, \quad (10)$$

provided $\nu \ll 1$. Application of the usual percolation theories suggests that infinite-size networks of false vacuum domains disappear when $u(\tau)$ becomes less than a certain value u_c (≈ 0.3) [13]^{†1}; hence all the false vacuum domains become surrounded by true vacuum bubbles. The critical time τ_c ($u(\tau_c) = u_c$), easily determined from eq. (9), is given approximately by

$$\tau_c - \tau_* \approx \nu^{-1} l. \quad (11)$$

The mean separation distance of bubbles at this time is about $d = n_B^{-1/3} \approx (10/\nu)^{1/3} l$. We may presume that the mean diameter of trapped false vacuum regions is of the same order, i.e.

$$r_-(\tau_c) = 0.5(10/\nu)^{1/3} l. \quad (12)$$

By comparing this equation with eq. (8), we find that, if the factor $(10/\nu)^{1/3}$ is greater than unity, most of the trapped false vacuum regions become wormholes eventually. Therefore, we can regard τ_c as the completion time of the phase transition [11]. As has been discussed in the literature [1–3], the duration of the phase transition era must be greater than $60 l$ in order to explain the homogeneity of the present universe by a GUT phase transition (as a consequence of the exponential expansion in the size of the particle horizon). If the nucleation rate ν is smaller than $1/60$, both this condition and the condition for the wormhole creation are satisfied.

The Einstein–Rosen bridge which connects a sub-universe (Region A in fig. 1) with the original universe will disappear and the sub-universe will become spatial-

ly disconnected^{†2} if wormholes evaporate as black holes do. In order to estimate the evaporation time t_{ev} , we need the value of the average mass of wormholes. From eqs. (5), (6), (8) and (12) we find the mass of a typical wormhole as

$$M_w \approx (k/8)(10/\nu)(4\pi\rho_v l^3/3), \quad (13)$$

where k is a numerical factor of the order $\nu/10 < k < 1$. Then the evaporation time t_{ev} is given as

$$t_{ev} = \frac{10\pi \times 8^4}{27} M^3 \approx 0.14 k^3 (10/\nu)^3 (m_p^4/\rho_v) l, \quad (14)$$

where m_p is the Planck mass ($\approx 10^{19}$ GeV). If we take a plausible model for the phase transition, $\rho_v = (10^{15} \text{ GeV})^4$ (GUT phase transition) and $\nu = 1/60$, t_{ev} is evaluated as

$$t_{ev} \approx 3 \times 10^{-13} k^3 \text{ (s)}. \quad (15)$$

Note that this corresponds to a cosmic time sufficiently before the Weinberg–Salam phase transition. The time t_{ev} in eq. (15) should be considered as the lapse of the Schwarzschild time appearing in eq. (4). The corresponding lapse of the de Sitter time is determined from the junction condition of the metrics (1) and (4) at the boundary, which yields a relation between the two time coordinates t and τ [11];

$$t \approx r_-(\tau_0) \exp[(\tau - \tau_0)/l]. \quad (16)$$

The phase transition proceeds also in the sub-universes. The completing time of the phase transition in these sub-universes is estimated as $\tau = \tau_0 + \nu^{-1} l$ from eq. (11), because the ignition time of the phase transition is now τ_0 instead of τ_* . The corresponding Schwarzschild time t_c is given by

$$t_c = r_-(\tau_0) \exp(\nu^{-1}) \lesssim 10^{26} l \approx 10^{-10} \text{ s}. \quad (17)$$

Comparison of this equation with eq. (15) shows $t_{ev} \ll t_c$, i.e. before the phase transition completes, the sub-universes connected by wormholes with the original universe become spatially disconnected entirely^{†3}. We call such a sub-universe a child universe with respect to the original “mother” universe.

^{†1} Readers may suspect that the percolation theory in flat space can be applied to cases with such a complicated geometry as the present one. However, we believe this application is valid for two reasons. One is that the theory is applied only to the false vacuum region which is spatially flat. The other is that the volume fraction [eq. (9)] is concerned with not the proper but the coordinate volume. We also note that the essential scenario of the present work does not depend on the actual value of u_c .

^{†2} We mean by “spatially disconnected” that there is a connected spacelike slice Σ of the universe such that it does not intersect with the causal boundary of the universe and its causal future, $J_+(\Sigma)$, is composed of topologically disconnected components.

^{†3} See next page.

The radius of a child universe immediately after the phase transition is given as

$$L \lesssim (10/\nu)^{1/3} \exp(\nu^{-1}) l. \quad (18)$$

For $\rho_v = (10^{15} \text{ GeV})^4$ and $\nu = 1/60$, L is about 10 cm. This length is large enough to regard our universe as such a child universe; since the universe expands 10^{28} times as large [the ratio of the GUT temperature (10^{15} GeV) to the present background radiation temperature (10^{-4} eV)] after the phase transition, the present radius of a child universe is larger than $H_0^{-1} \approx 10^{28} \text{ cm}$ (H_0 is the Hubble constant). Thus there are no observational conflicts.

One easily arrives at an idea that wormholes are further created by the phase transition in the child universes and they also evaporate to produce grandchild universes. This sequential production of universes may continue on and on (see fig. 2). Now a very peculiar consequence of a cosmological first-order phase transition is at hand; although the Creator might have made a unitary universe, the universe itself is also capable of bearing child universes, which are again capable of bearing universes, and so on. Our universe is too old to

bear more child universes at present, though it might have born them during the period of the phase transition or/and it might have been born as one of the child universes in this sequence.

How many child universes can one universe bear? The number density of false vacuum domains when percolating networks of false vacua disappear is estimated from eqs. (9) and (12) as $n_f \approx u_c/[4\pi r_-(\tau_c)^3/3] = \nu/(30l^3)$. The volume of each spatially disconnected universe at this instance is $4\pi L^3/3$. Thus, if all the false vacuum domains become wormholes, the production rate of child universes per universe is estimated as

$$p = n_f(4\pi L^3/3) \approx \exp(3/\nu)/3. \quad (19)$$

For $\nu = 1/60$, eq. (19) yields an extremely huge number 10^{77} !

In summary, we have shown that gauge theories in current particle physics predict multi-production of universes if the phase transition is of first-order. There are, however, a few problems yet to be solved. One is on the thermalization mechanism of the kinetic energy of bubble walls at their mutual collisions. In particular, if the size of a bubble is much greater than l , it seems unlikely that the kinetic energy is transformed to radiation quickly. Another is on the baryon-entropy ratio. When black holes and wormholes evaporate, a large amount of entropy is generated, which consequently reduces the baryon number to entropy ratio considerably. Thus, even if the ratio is initially of order unity, the evaporation may dilute the baryon excess too much to agree with observations. In order to avoid this difficulty, some new baryon-number generation mechanism might be necessary [15]. To judge the validity of the present result, more precise investigations based on the further development of particle physics will be required.

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^{‡3} Very recently, we have discussed the fate of wormholes in detail taking into account the nucleation of bubbles in a trapped false vacuum region. The result indicates that a wormhole once formed either disappears leaving no singularity or collapses to a black hole if the phase transition in the trapped region completes before this region is disconnected by the wormhole evaporation. Thus the condition $t_{ev} \ll t_c$ may be indispensable in order for these subuniverses to survive. See for details ref. [14].

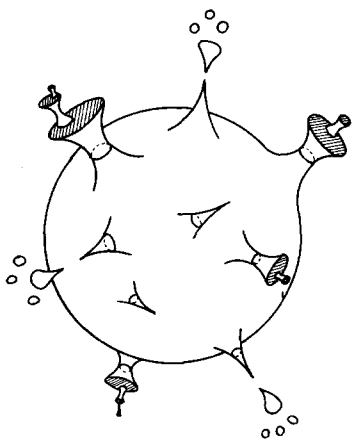


Fig. 2.

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